

Conservation of Momentum Simulation

Purpose: The purpose of this lab is to observe conservation of momentum in collisions.

Background: Momentum is mass (inertia) in motion. The mathematical equation is: $p = mv$, where p is momentum, m is mass and v is velocity. Just like velocity, momentum is a vector quantity that has both magnitude and direction.

In this simulation you will learn about the fundamental principle of physics called conservation of momentum which states that the momentum of a system remains constant in the absence of external forces. In this activity, you will be dealing with a simple **system of two balls** that collide with each other. For a collision between two objects, the total momentum before the collision is equal to the total momentum after the collision.

Elastic Collisions: defined as one in which there is no loss of kinetic energy

Inelastic Collisions: defined as one in which part of the kinetic energy is converted into some other form

Instructions:

Go to https://phet.colorado.edu/sims/html/collision-lab/latest/collision-lab_en.html

Click on the Explore 1D module.

*Note: It is helpful to run all simulations in slow motion.

Part 1: Elastic Collisions

Simulation 1

Click on *More Data*. Set the mass of both balls to 1.0 kg each. For ball 1, set the velocity to 1 m/s. Set the **velocity of ball 2 to zero**. Leave the positions as the defaults. In the gray box, click on *Momentum*, which shows the momentum vectors for each ball. Also, click on *Values* and *Kinetic Energy*. Set the **elasticity** to 100%. Make sure “REFLECTIVE BORDER” is OFF!! Click on the play button. Fill in the data below for before and after the collision. To determine the change in momentum for each ball subtract its momentum before the collision from the its momentum after the collision. (Don’t forget momentum is a vector quantity.)

	Momentum Before (kg m/s)	Momentum After (kg m/s)	Change in Momentum (kg m/s)	Velocity Before (m/s)	Velocity After (m/s)	Kinetic Energy Before (J)	Kinetic Energy After (J)
Ball 1							
Ball 2	0			0			
Total System							

1. What happens to ball 1 after it strikes ball 2?

- Was momentum conserved in this simple system of two billiard balls?
- Did the kinetic energy of the system change? Why is this?

Simulation 2

Reset the sim. Now, set the mass of ball 1 to 0.5 kg and ball 2 to 1.5 kg. **Keep all other parameters the same as in simulation 1 (start ball 2 at rest).** Make sure “REFLECTIVE BORDER” is OFF!! Click the play button. Record data below. (Keep in mind that momentum is a vector, so pay attention to positive and negative values.)

	Momentum Before (kg m/s)	Momentum After (kg m/s)	Change in Momentum (kg m/s)	Velocity Before (m/s)	Velocity After (m/s)	Kinetic Energy Before (J)	Kinetic Energy After (J)
Ball 1							
Ball 2	0			0			
Total System							

- Describe what happens once ball 1 strikes ball 2.
- Why are the momentum and velocity of ball 1 negative numbers?
- Was momentum conserved for the system of two balls?
- Did the kinetic energy of the system change? Why is this?

Simulation 3

Reset the sim. Keep the mass of ball 1 at 0.5 kg and always start it with a velocity of 1 m/s. Change the mass of ball 2 to the values listed in the table below and **always start it at rest.** Maintain 100% elasticity. Make sure “REFLECTIVE BORDER” is OFF!! Run the simulation in each case and fill in the table. BE SURE TO RESET AFTER EACH RUN!

Ball 2 Mass (kg)	Total momentum Before (kg m/s)	Momentum of Ball 1 after (kg m/s)	Momentum of Ball 2 after (kg m/s)	Total Momentum After (kg m/s)	Velocity of Ball 1 after (m/s)	Velocity of Ball 2 after (m/s)
0.5						
1.0						
1.5						
2.0						
2.5						
3.0						

8. With an increase in the mass of Ball 2, the velocity of Ball 2 _____ after the collision.
 - a. increases
 - b. decreases

9. With an increase in the mass of Ball 2, the speed and momentum (magnitude only) of Ball 1 _____ after the collision
 - a. increase
 - b. decrease

Unfortunately, in this simulation, we are unable to increase the mass of ball 2 beyond 3 kg. Based upon the trends of data in the table above, answer the following question.

10. Imagine Ball 2 is the Earth (6×10^{24} kg). Extend the situation to an elastic collision with a ball bouncing off a rigid floor (the Earth). The magnitudes of momentum and velocity of Ball 1 after the collision should be _____ the momentum and velocity magnitudes just before the collision.
 - a. essentially equal to
 - b. more than
 - c. less than

Part 2: Inelastic Collisions

Simulation 4

Reset the simulation. Set both masses to 1.5 kg. Set the elasticity to 0% (**inelastic**). Set the velocity of ball 1 to 1 m/s and the velocity of ball 2 to zero. Make sure “REFLECTIVE BORDER” is OFF!! Run the sim and fill in the table.

	Momentum Before (kg m/s)	Momentum After (kg m/s)	Change in Momentum (kg m/s)	Velocity Before (m/s)	Velocity After (m/s)	Kinetic Energy Before (J)	Kinetic Energy After (J)
Ball 1							
Ball 2	0			0			
Total System							

Reset the simulation. Set the mass of ball 1 to 1.5 kg and the mass of ball 2 to 3 kg. Set the elasticity to 0% (**inelastic**). Set the velocity of ball 1 to 1 m/s and the velocity of ball 2 to zero. Make sure “REFLECTIVE BORDER” is OFF!! Run the sim and fill in the table.

	Momentum Before (kg m/s)	Momentum After (kg m/s)	Change in Momentum (kg m/s)	Velocity Before (m/s)	Velocity After (m/s)	Kinetic Energy Before (J)	Kinetic Energy After (J)
Ball 1							
Ball 2	0			0			
Total System							

11. Describe the motion of the two-ball system after both of these inelastic collisions.

12. For both of the above scenarios, was total momentum conserved?

13. Did the kinetic energy of the system change? Why is this?

Closing Questions:

14. Explain how conservation of momentum relates to Newton's third law of motion. (Hint: the concept of impulse is important here.)

15. Picture two astronauts holding onto one another in space. Their initial velocity with respect to each other is zero.
 - a. If one astronaut pushes the other away, what is the total momentum of both astronauts combined?

 - b. If one astronaut has twice as much mass as the other, what can you say about the velocity of the less massive one compared to the other after they push off each other?